Research Presentation

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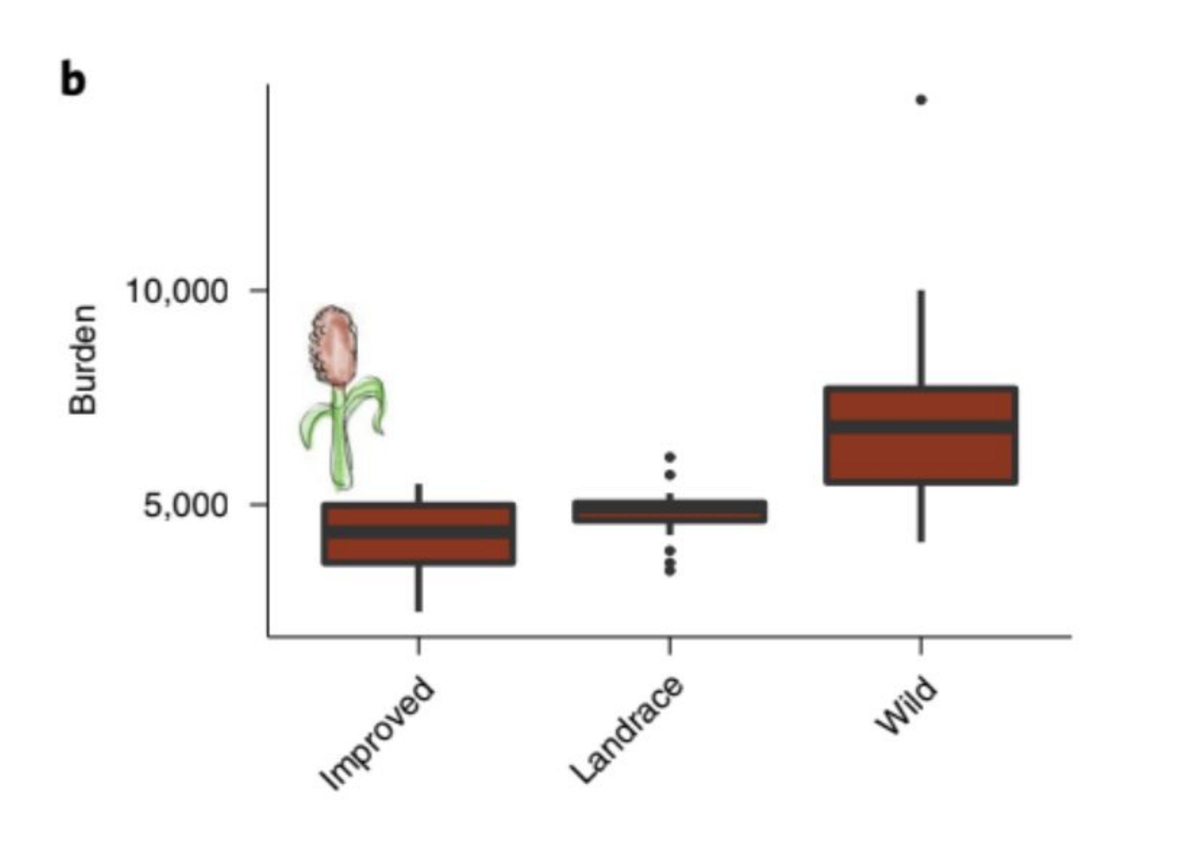
Research Question

Based on what methods you use to determine the ancestral allele, what results do you get?

The Lozano Paper



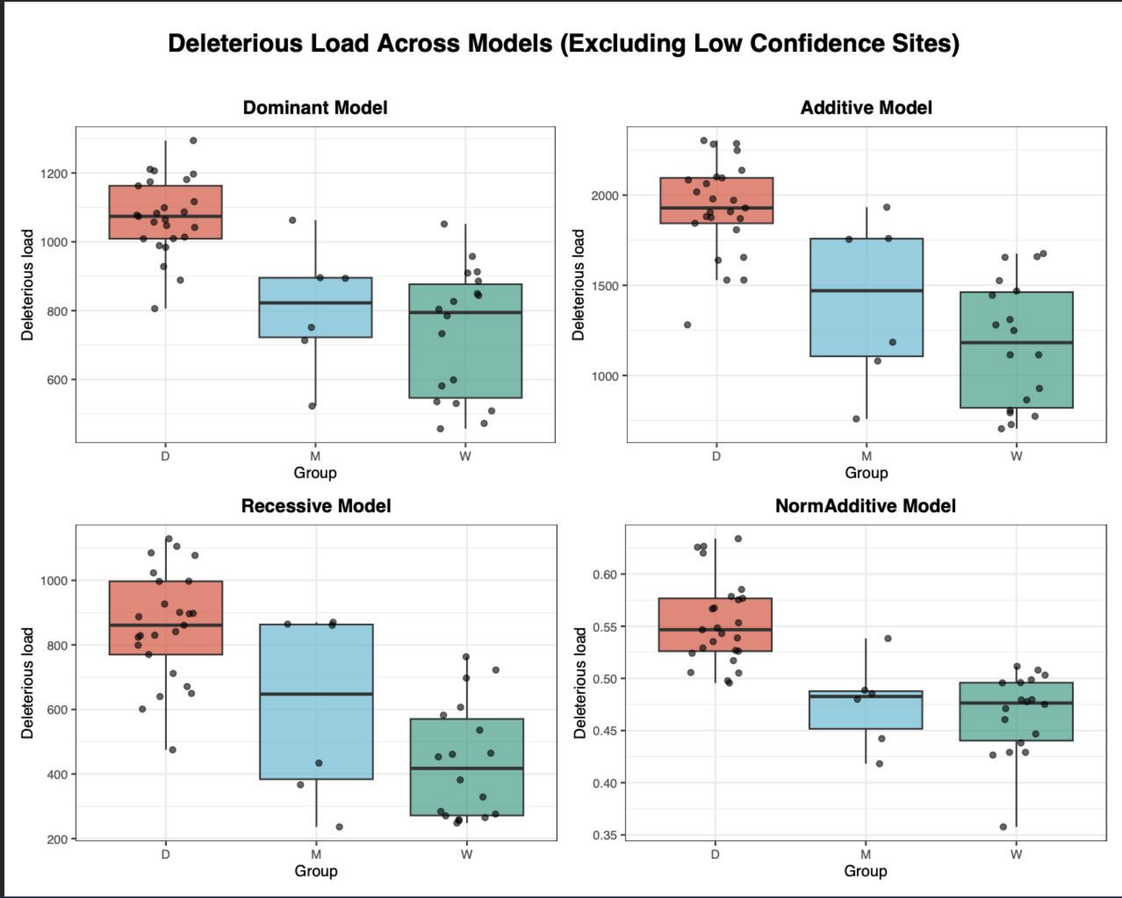
https://doi.org/10.1038/s41477-020-00834-5

Polarization Methods

01. **Use outgroup as ancestral**

02. **Lozano paper pipeline**

Using the outgroup as the ancestral allele



Polarization Methods

01. **Use outgroup as ancestral**

02. **Lozano paper pipeline**

The Lozano Paper Pipeline

01. **Orthogroups**

02. **Clustal Omega**

03. **TranAlign**

04. **Mapping**

05. **est-sfs software**

Straightforward...or is it?

CSV/TSV files

CDS FASTAs

Nucleotide
alignments

Text files

VCF files

GFF3 files

Protein alignments

Protein FASTAs

Step 0: Organizing Data
Which files do I need?

Focal species: Sorghum bicolor (BTx623)

- Protein FASTA
- CDS FASTA
- Gff3 file
- VCF file

Outgroups: Setaria italica, Maize

- Protein FASTAs
- CDS FASTAs

Step 0: Organizing Data
Deal Breakers

- Protein FASTAs for each species→NO ASTERISK (TERMINAL CODON)

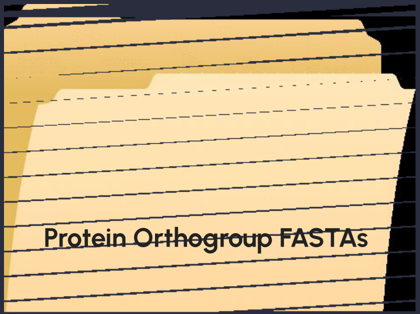
- ***SEQUENCE HEADERS MUST MATCH IN BOTH PROTEIN AND CDS FASTAS***
  - >Zm00001d033527_P001
  - >Zm00001d033527_T001
  - Trailing ‘.p’ → remove it!!!

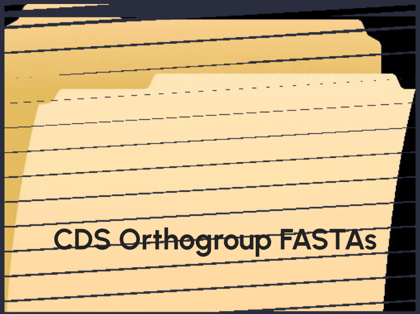
Step 1: Orthogroups

OrthoFinder

Focal species: Sorghum bicolor (BTx623)

Outgroups: Setaria italica, Maize

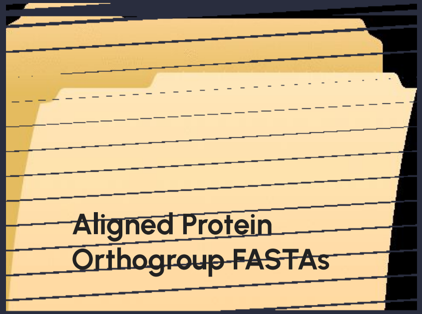




Step 2: Protein Orthogroup Alignments

Clustal Omega

- Go through each protein orthogroup FASTA and align each one with Clustal Omega



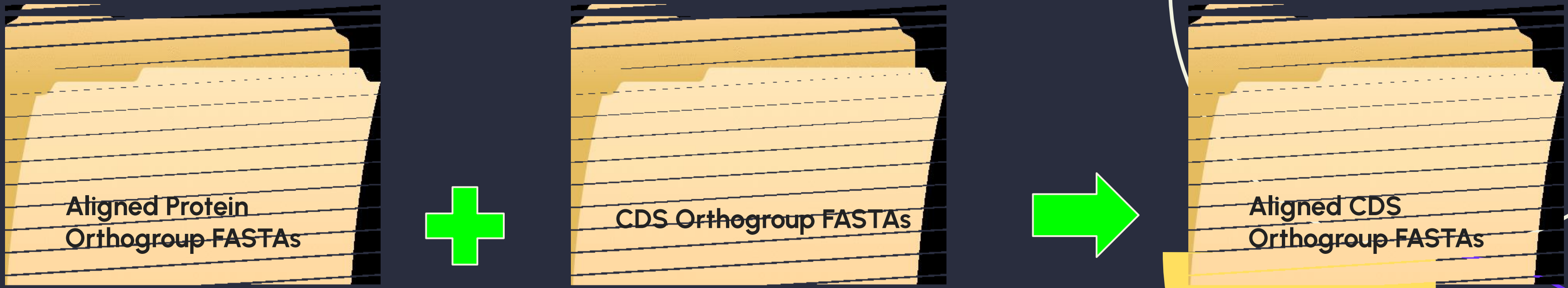
Step 3: Nucleotide Orthogroup Alignment

TranAlign

- Part of the EMBOSS package
- Requires an unaligned nucleotide fasta and an aligned protein fasta

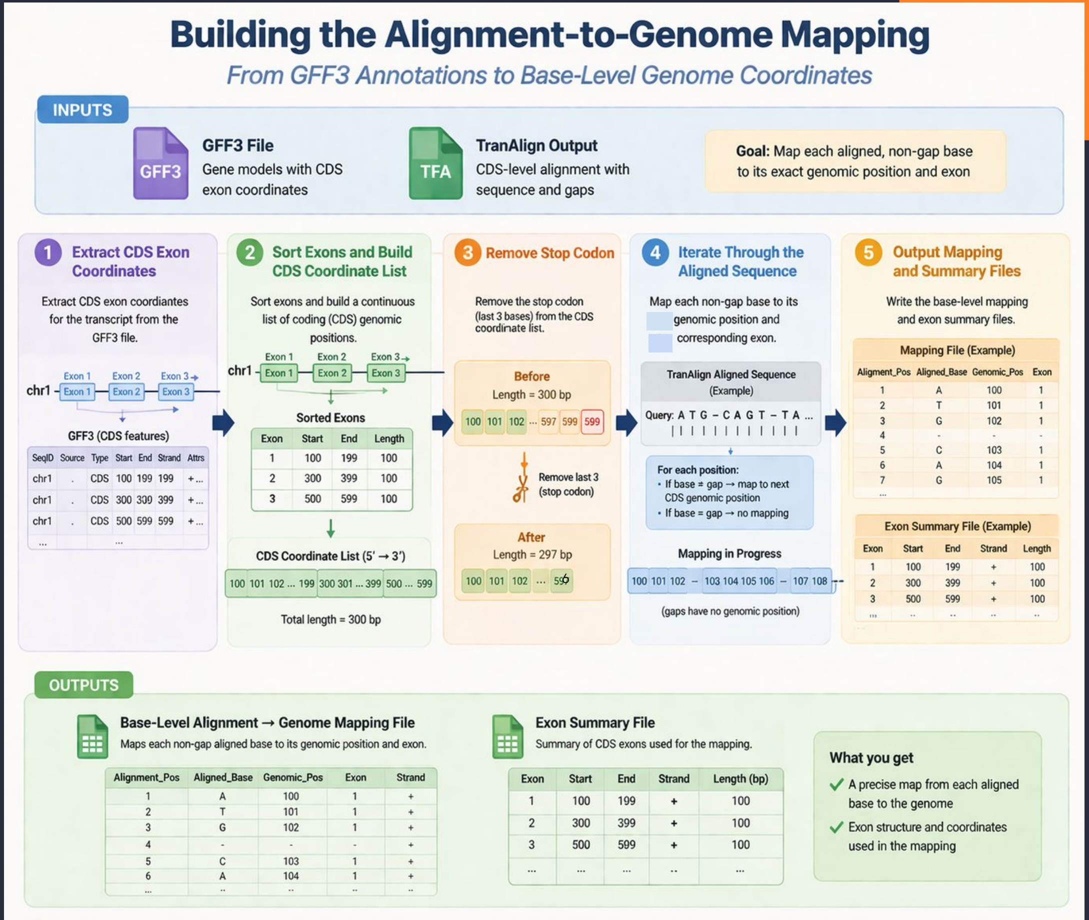
Step 3: Nucleotide Orthogroup Alignment

TranAlign



Step 4: Mapping

Translating alignment positions to their position in the genome



Step 4: Mapping

Plus Strand Example

```
Exon_Number        Start    End       Strand
1          5360374 5360638 +
2          5360780 5361237 +
3          5361362 5361745 +
Sobic.004G066000_exons.tsv (END)
```

```
[gjacks21@str-i3 mapped_coordinates]$ head Sobic.004G066000_mapping.csv
Base,Alignment_Position,Genomic_Position,Exon
A,1,5360374.0,1.0
T,2,5360375.0,1.0
G,3,5360376.0,1.0
G,4,5360377.0,1.0
C,5,5360378.0,1.0
T,6,5360379.0,1.0
G,7,5360380.0,1.0
G,8,5360381.0,1.0
C,9,5360382.0,1.0
[gjacks21@str-i3 mapped_coordinates]$ tail Sobic.004G066000_mapping.csv
A,1248,5361733.0,3.0
C,1249,5361734.0,3.0
A,1250,5361735.0,3.0
G,1251,5361736.0,3.0
G,1252,5361737.0,3.0
G,1253,5361738.0,3.0
A,1254,5361739.0,3.0
C,1255,5361740.0,3.0
A,1256,5361741.0,3.0
G,1257,5361742.0,3.0
```

Step 4: Mapping

Plus Strand Example: Zooming in on Exons

Exon_Number	Start	End	Strand
1	5360374	5360638	+
2	5360780	5361237	+
3	5361362	5361745	+

Sobic.004G066000_exons.tsv (END)

```
-,376,,
-,377,,
-,378,,
T,379,5360635.0,1.0
C,380,5360636.0,1.0
A,381,5360637.0,1.0
G,382,5360638.0,1.0
G,383,5360780.0,2.0
A,384,5360781.0,2.0
T,385,5360782.0,2.0
G,386,5360783.0,2.0
```

```
C,874,5361235.0,2.0
A,875,5361236.0,2.0
G,876,5361237.0,2.0
T,877,5361362.0,3.0
G,878,5361363.0,3.0
```

Step 4: Mapping

Minus Strand Example

Exon_Number	Start	End	Strand
1	61354977	61355051	-
2	61354009	61354212	-
3	61353227	61353484	-
4	61352712	61352927	-
5	61352471	61352626	-
6	61352228	61352387	-
7	61351988	61352139	-
8	61351726	61351813	-
9	61351577	61351650	-
10	61351422	61351496	-
11	61351002	61351334	-

Sobic.002G221700_exons.tsv (END)

```
[gjacks21@str-i3 mapped_coordinates]$ head Sobic.002G221700_mapping.csv
Base,Alignment_Position,Genomic_Position,Exon
A,1,61355051.0,1.0
T,2,61355050.0,1.0
G,3,61355049.0,1.0
G,4,61355048.0,1.0
T,5,61355047.0,1.0
G,6,61355046.0,1.0
G,7,61355045.0,1.0
G,8,61355044.0,1.0
C,9,61355043.0,1.0
[gjacks21@str-i3 mapped_coordinates]$ tail Sobic.002G221700_mapping.csv
C,1920,61351014.0,11.0
T,1921,61351013.0,11.0
C,1922,61351012.0,11.0
T,1923,61351011.0,11.0
A,1924,61351010.0,11.0
T,1925,61351009.0,11.0
T,1926,61351008.0,11.0
G,1927,61351007.0,11.0
G,1928,61351006.0,11.0
A,1929,61351005.0,11.0
```

Step 4: Mapping

Minus Strand Example: Zooming in on Exons

Exon_Number	Start	End	Strand
1	61354977	61355051	-
2	61354009	61354212	-
3	61353227	61353484	-
4	61352712	61352927	-
5	61352471	61352626	-
6	61352228	61352387	-
7	61351988	61352139	-
8	61351726	61351813	-
9	61351577	61351650	-
10	61351422	61351496	-
11	61351002	61351334	-

Sobic.002G221700_exons.tsv (END)

```
A,122,61354978.0,1.0
G,123,61354977.0,1.0
C,124,61354212.0,2.0
A,125,61354211.0,2.0
A,126,61354210.0,2.0
```

```
A,326,61354010.0,2.0
G,327,61354009.0,2.0
A,328,61353484.0,3.0
A,329,61353483.0,3.0
A,330,61353482.0,3.0
```

Step 5: est-sfs software

Estimated unfolded site frequency spectrum software

- Input files:
  - config-file.txt
  - seed-file.txt
  - **data-file.txt**

Step 5: est-sfs software

config-file.txt

5.1. Configuration file (config-file.txt)

This file (examples provided) consists of three lines of text in the following order:

n_outgroup [1, 2, or 3]
model [0, 1 or 2]
nrandom [0 or positive integer]

<u>model</u>

0 = Jukes-Cantor model
1 = Kimura 2-parameter model
2 = Rate-6 model (see Keightley and Jackson 2018 for details)

<u>nrandom</u>

If nrandom = 0, the ML search algorithm starts with pre-set parameter starting values. If nrandom >= 1, the algorithm returns the highest log likelihood found in nrandom runs using starting values for the parameters randomly sampled from wide ranges. The MLs of the individual runs are reported. In the case of the Rate-6 model, it is recommended that at least 10 random starting value runs are carried out to ensure convergence.

Source: est-sfs software documentation

Step 5: est-sfs software
seed-file.txt

5.3. Seed file (**seed-file.txt**)

This text file (example provided) contains a single positive integer. It is overwritten by a new random number seed at the end of the run.

Source: est-sfs software documentation

Step 5: est-sfs software
data-file.txt

If there are three outgroups, there are 4 space-separated columns. The first column is for the focal species, and the next three columns are for the outgroups. Each column is a comma-separated list of the counts of the four bases in the order A, C, G, T. For example, the first line in the example data file is:

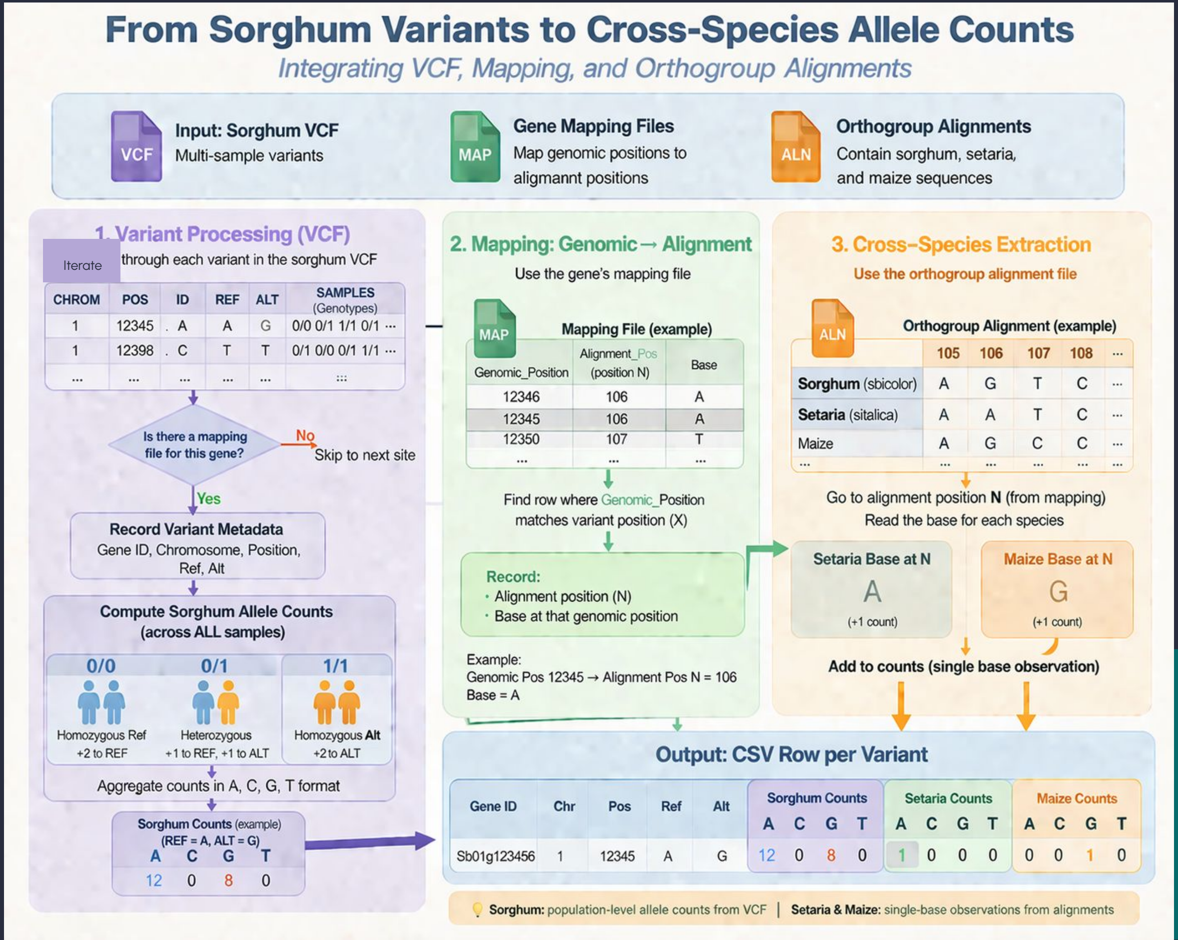
```
20,0,0,0        0,0,0,1 0,0,0,1 0,0,0,1
```

At this site, all $n = 20$ copies sampled in the focal species are A. In the three outgroups, a single copy has been sampled, and in each case it is T. All sites must have the same number of copies sampled in the focal species and up to one copy sampled in each outgroup. If there are missing data in any outgroup, the counts for that outgroup are encoded `0,0,0,0` . Data from polymorphic and non-polymorphic sites are analysed together.

Source: est-sfs software documentation

Step 5: est-sfs software

Generating data-file.txt



Step 5: The Program
est-sfs software

- Input files:
  - config-file.txt
  - seed-file.txt
  - **<u>data-file.txt</u>**
- Output files
  - output-file-sfs.txt
  - **<u>output-file-pvalues.txt</u>**

Step 5: est-sfs software

Results

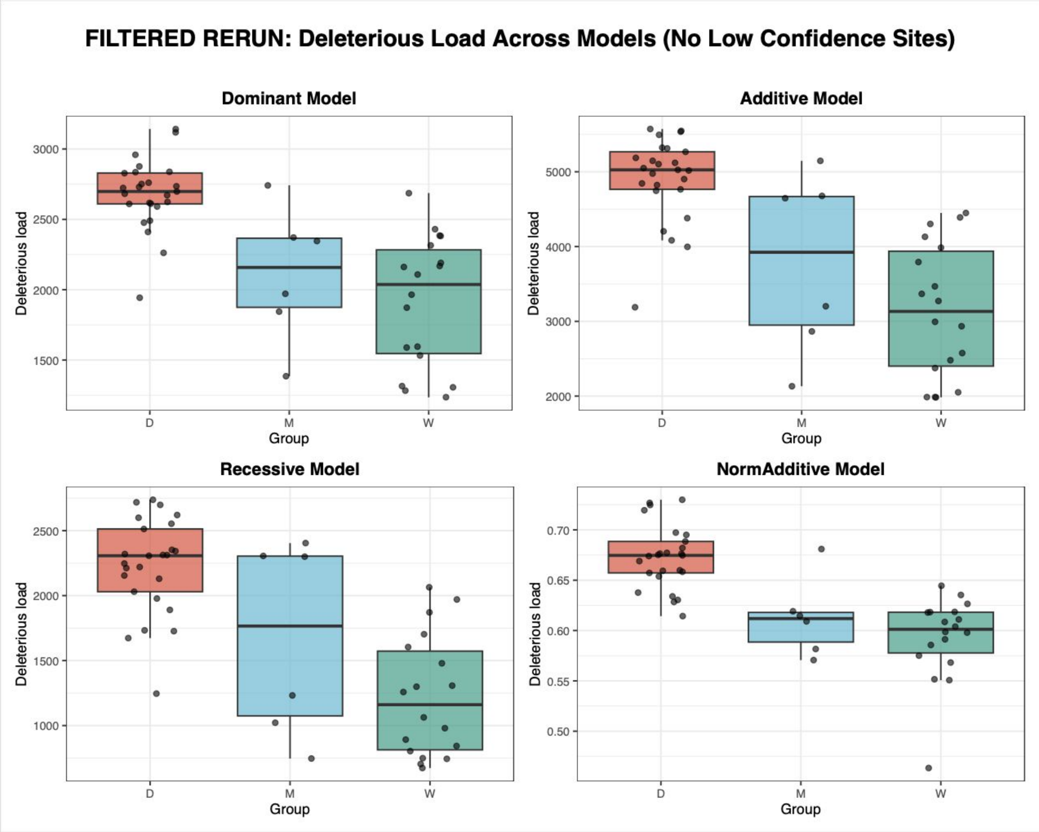
A	B	C	D	E	F	G	H	I	J	K	L	M	N
GeneID	Chromosome	GenomicPos	Ref	Alt	Sorghum Counts	Setaria Counts	Maize Counts	est-sfs probability	Major_Allele	Minor_Allele	est_Ancestral_Allele	est_Ancestral_Confidence	Propq1 Ancestral_Allele
Sobic.001G000900	Chr01	80269	T	C	A:0 C:12 G:0 T:76	A:0 C:0 G:0 T:2	A:0 C:0 G:0 T:2	0.985385	T	C	T	high	T
Sobic.001G000900	Chr01	80344	T	C	A:0 C:8 G:0 T:80	A:0 C:0 G:0 T:0	A:0 C:2 G:0 T:0	0.547193	T	C	T	low	T
Sobic.001G000900	Chr01	80439	C	T	A:0 C:79 G:0 T:9	A:0 C:2 G:0 T:0	A:0 C:2 G:0 T:0	0.981855	C	T	C	high	C
Sobic.001G000900	Chr01	80703	G	A	A:3 C:0 G:81 T:0	A:0 C:0 G:0 T:0	A:0 C:0 G:0 T:0	0.876038	G	A	G	moderate	G
Sobic.001G000900	Chr01	81102	G	A	A:3 C:0 G:87 T:0	A:0 C:2 G:0 T:0	A:0 C:0 G:2 T:0	1	G	A	G	high	G
Sobic.001G000900	Chr01	81103	C	A	A:3 C:87 G:0 T:0	A:0 C:2 G:0 T:0	A:0 C:2 G:0 T:0	1	C	A	C	high	C
Sobic.001G000900	Chr01	81348	G	T	A:0 C:0 G:78 T:12	A:0 C:0 G:0 T:0	A:0 C:0 G:2 T:0	0.969255	G	T	G	high	G
Sobic.001G003500	Chr01	305187	C	G	A:0 C:79 G:13 T:0	A:0 C:2 G:0 T:0	A:0 C:4 G:0 T:0	0.981855	C	G	C	high	C
Sobic.001G008200	Chr01	759670	A	T	A:61 C:0 G:0 T:21	A:0 C:2 G:0 T:0	A:0 C:2 G:0 T:0	0.699708	A	T	A	moderate	T
Sobic.001G008200	Chr01	760099	C	G	A:0 C:63 G:19 T:0	A:0 C:0 G:2 T:0	A:0 C:0 G:0 T:2	0.454372	C	G	G	low	G
Sobic.001G008200	Chr01	761129	A	G	A:73 C:0 G:5 T:0	A:2 C:0 G:0 T:0	A:2 C:0 G:0 T:0	0.930911	A	G	A	high	A
Sobic.001G008200	Chr01	761800	G	A	A:6 C:0 G:78 T:0	A:0 C:0 G:2 T:0	A:0 C:0 G:2 T:0	0.973666	G	A	G	high	G
Sobic.001G008200	Chr01	762219	T	A	A:6 C:0 G:0 T:74	A:0 C:0 G:0 T:2	A:0 C:2 G:0 T:0	0.85484	T	A	T	moderate	T

Step 5: est-sfs software

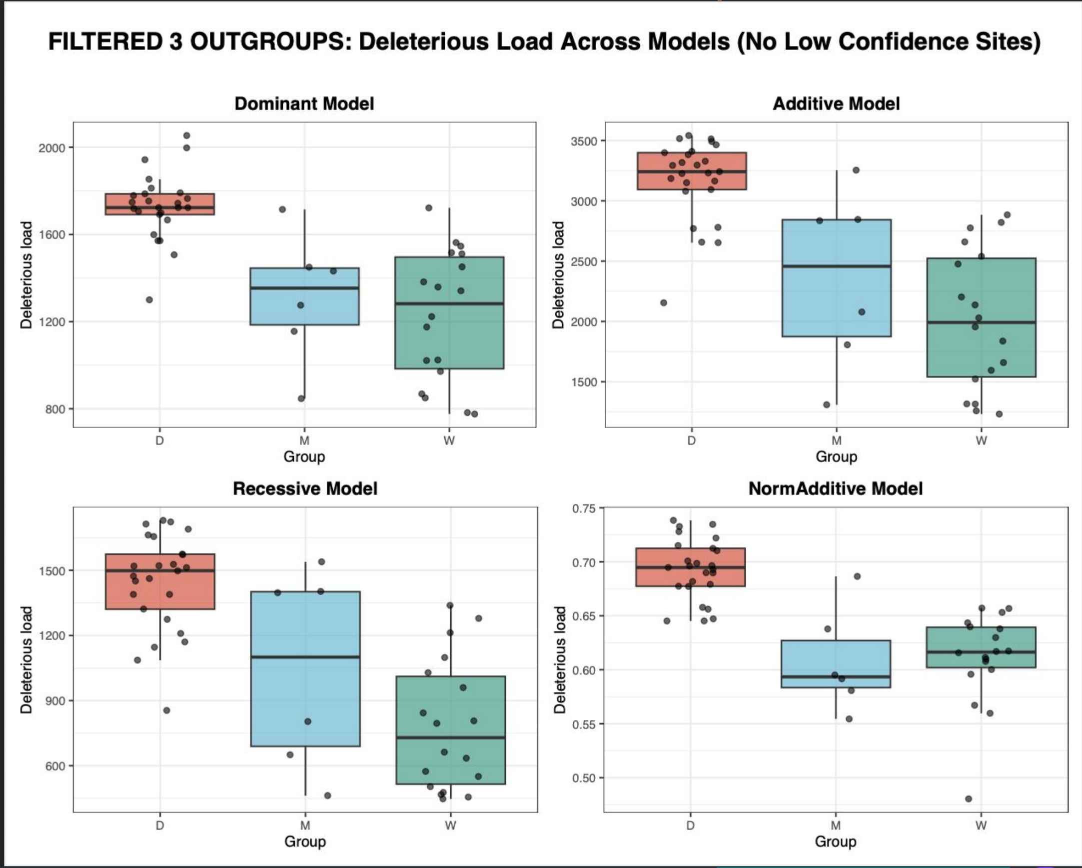
Results

```
(base) gloriajackson@glorias-MacBook-Pro replicating_lozano % python compare_ancestral_alleles.py
=== Ancestral allele comparison ===
Variants compared : 7673
Matches           : 5697
Mismatches        : 1976
Agreement (%)     : 74.247%
```

Step 5: est-sfs software
Results



Step 5: est-sfs software
3 Outgroups

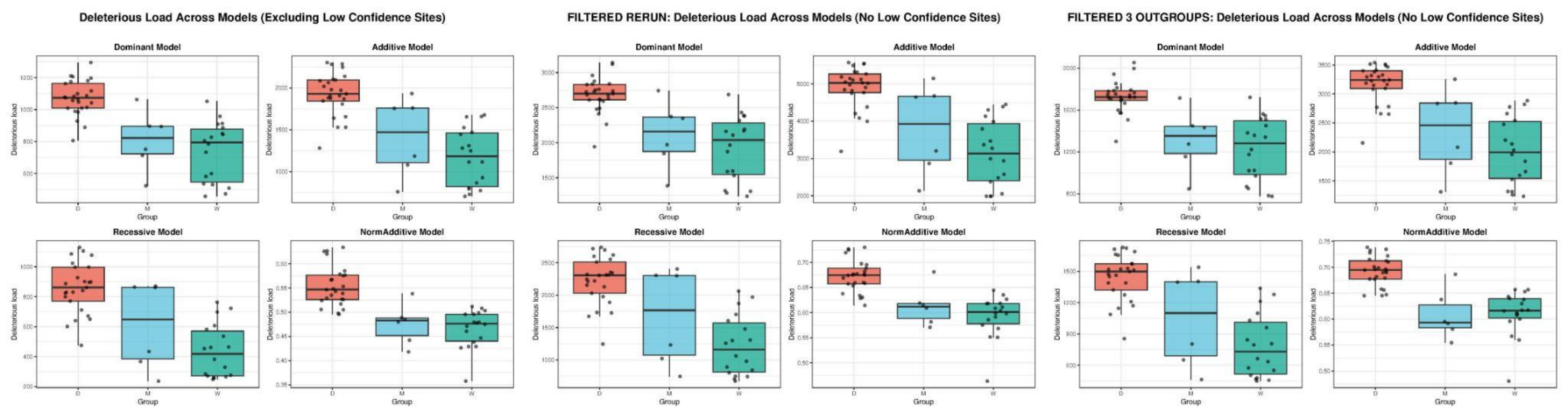


Step 5: est-sfs software

Comparing Results

```
(base) gloriajackson@glorias-MacBook-Pro replicating_lozano % less compare_ancestral_calls.py
(base) gloriajackson@glorias-MacBook-Pro replicating_lozano % python compare_ancestral_calls.py
Total shared variants: 3453
Matching variants: 3154
Disagreement variants: 299
(base) gloriajackson@glorias-MacBook-Pro replicating_lozano %
```

Put it all together now!



Takeaways

- Went into this thinking polarization might have a big impact on results…it doesn't

- Deleterious load consistently higher in domesticated regardless of polarization method

Thank you :)